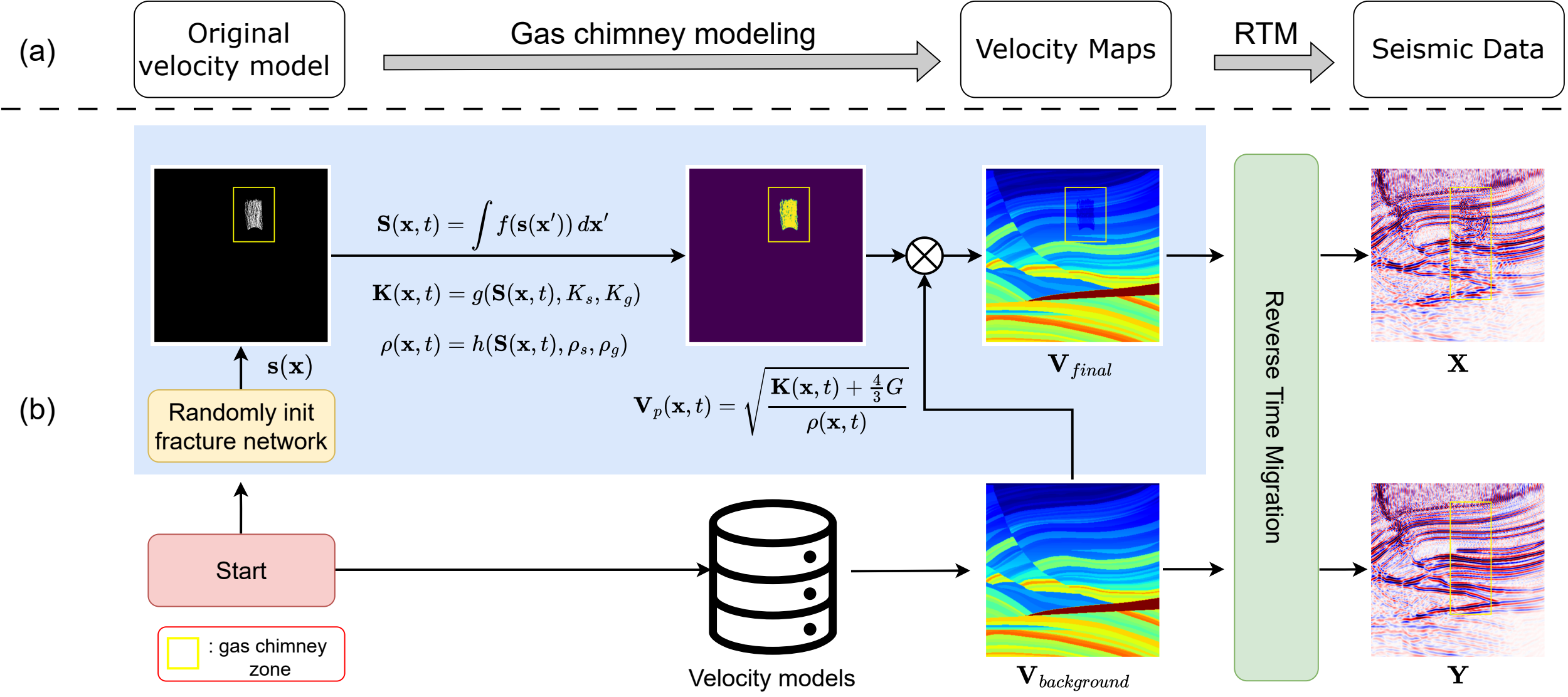
(a)
Original velocity model
Gas chimney modeling
Velocity Maps
RTM
Seismic Data
(b)
$\mathbf{S}(\mathbf{x},t) = \int f(\mathbf{s}(\mathbf{x}'))\, d\mathbf{x}'$
$\mathbf{K}(\mathbf{x},t) = g(\mathbf{S}(\mathbf{x},t), K_s, K_g)$
$\rho(\mathbf{x},t) = h(\mathbf{S}(\mathbf{x},t), \rho_s, \rho_g)$
$\mathbf{V}_p(\mathbf{x},t) = \sqrt{\frac{\mathbf{K}(\mathbf{x},t) + \frac{4}{3}G}{\rho(\mathbf{x},t)}}$
$\mathbf{s}(\mathbf{x})$
Randomly init fracture network
$\mathbf{V}_{final}$
Reverse Time Migration
$\mathbf{X}$
Start
Velocity models
$\mathbf{V}_{background}$
$\mathbf{Y}$
: gas chimney zone